Resilience Academy

SDI System Installation & Customization with CRD and KoboToolbox

Session 1.1: Introduction to SDI Concepts

What is a Spatial Data Infrastructure?

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Day 1 | Lecture 1 (30 min) | crd.resilienceacademy.ac.tz

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Session Overview

Goal: Understand what an SDI is and how the Climate Risk Database fits in

By end of this session you can explain the five pillars of an SDI, name four OGC standards, and navigate the live CRD platform.

Topics

- What is Spatial Data?
- Vector vs Raster data types
- Coordinate Reference Systems & formats
- What is an SDI?
- Five pillars: data, metadata, services, standards, people
- OGC standards: WMS, WFS, WCS, CSW
- CRD as a GeoNode-based SDI

Activities

- Live demo: crd.resilienceacademy.ac.tz
- Search exercise: find Tanga & Mwanza datasets
- Q&A discussion

Reference Platform

`crd.resilienceacademy.ac.tz`

What is Spatial Data?

Definition

Spatial data (geospatial data) is any data that has a **geographic location** attached to it — it describes objects, events or phenomena on the Earth's surface.

Every spatial feature has:

- **Geometry** — where is it? (coordinates)
- **Attributes** — what is it? (name, type, population...)
- **Coordinate Reference System** — how are coordinates defined? (WGS 84, UTM...)
- **Temporal dimension** — when? (date

Examples of Spatial Data

Domain	Spatial Data
Climate risk	Flood extent polygons
Infrastructure	School/hospital locations
Business	Mwanza business points
Remote sensing	Tanga satellite imagery
Transport	Road network lines
Census	Ward population polygons
Hydrology	River lines, watersheds
Elevation	Digital Elevation Models

Key Insight

80% of all data has a spatial component. If it

Two Types of Spatial Data: Vector & Raster

Vector Data

Represents features as **discrete objects** using coordinates.

- **Points** — locations (schools, businesses, GPS readings)
  Example: `mwanza_businesses` — each business = one point
- **Lines** — linear features (roads, rivers, pipelines)
- **Polygons** — areas (wards, flood zones, buildings)
  Example: Tanga ward boundaries

Formats: GeoJSON, Shapefile (.shp), GeoPackage

Raster Data

Represents the world as a **grid of cells** (pixels), each with a value.

- **Satellite imagery** — RGB or multispectral bands
  Example: `Tanga_imagery_reduced` — 50cm KOMPSAT-3
- **DEMs** — elevation value per pixel
- **Land cover** — classification per pixel
- **Temperature / rainfall** — continuous surfaces

Formats: GeoTIFF (.tif), NetCDF, JPEG2000, Cloud Optimised GeoTIFF (COG)

Coordinate Reference Systems & Data Formats

Coordinate Reference Systems (CRS)

A CRS defines **how coordinates map to real locations** on Earth.

- **Geographic CRS** — latitude/longitude (degrees)
  WGS 84 (EPSG:4326) — used by GPS, CRD, Google Maps
  Tanga: lat −5.07, lon 39.10
  Mwanza: lat −2.50, lon 32.90
- **Projected CRS** — metres on a flat map
  UTM Zone 37S (EPSG:32737) — Tanzania
  Used for distance/area calculations

Wrong CRS = data appears in the wrong place!

Common Spatial Data Formats

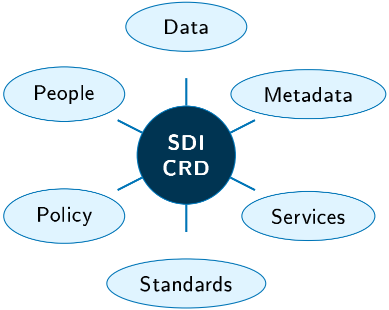
Format	Type	Used For
GeoJSON	Vector	Web APIs, CRD
Shapefile	Vector	Desktop GIS
GeoPackage	Vector	Modern standard
KML	Vector	Google Earth
GeoTIFF	Raster	Satellite imagery
COG	Raster	Cloud-optimised
NetCDF	Raster	Climate models
PostGIS	Both	Database storage

CRD Default

Upload in **any format** — CRD converts to

What is a Spatial Data Infrastructure?

A **framework** of policies, standards, technologies and people enabling geospatial data sharing and discovery.



The six pillars of SDI

Why SDI Matters

- Eliminates data duplication across institutions
- Enables interoperability between systems
- Supports evidence-based decision making
- Critical for climate risk management in Tanzania

Think of SDI as a **highway system** for geospatial data — the infrastructure that allows data to flow between producers and users.

The Five Pillars of an SDI

1. Data

Vector layers (points, lines, polygons) and raster imagery. Example: `mwanza_businesses` (point), `Tanga_imagery_reduced` (raster 50cm KOMPSAT-3).

2. Metadata

ISO 19115 standard descriptions — title, abstract, keywords, bounding box, date, contact. Enables search and discovery.

3. Services

4. Standards

- **OGC** — Open Geospatial Consortium
- **INSPIRE** — EU SDI directive
- **ISO 19100** series — geospatial metadata
- Ensures interoperability between platforms

5. People & Policy

Governance, data sharing agreements, institutional roles. CRD partners: UTU, Ardhi, SUZA, SUA, GIZ.

OGC Standards: The Language of SDI

Standard	Full Name	What It Does
WMS	Web Map Service	Returns **rendered map images** (PNG/JPEG). Used for basemaps and overlays.
WFS	Web Feature Service	Returns **raw vector data** (GeoJSON/GML). Enables download and analysis.
WCS	Web Coverage Service	Returns **raw raster data** (GeoTIFF). Used for satellite imagery and DEMs.
WMTS	Web Map Tile Service	Returns **pre-rendered tiles** for fast map display at multiple zoom levels.
CSW	Catalogue Service	**Search and discover** datasets

CRD: Tanzania's Climate Risk SDI

What is CRD?

- **Climate Risk Database** — Tanzania's national climate risk SDI
- Built on **GeoNode** open-source platform
- Hosts **287+ datasets**: flood maps, infrastructure, exposure, satellite imagery
- Partnership: Resilience Academy, UTU, Ardhi, SUZA, SUA, GIZ

Available Datasets

Region	Example Layers
Mwanza	mwanza_businesses
	mwanza_offices
	mwanza_education_facilities
Tanga	Tanga_imagery_reduced
	Tanga flood mapping layers
National	287+ climate risk datasets

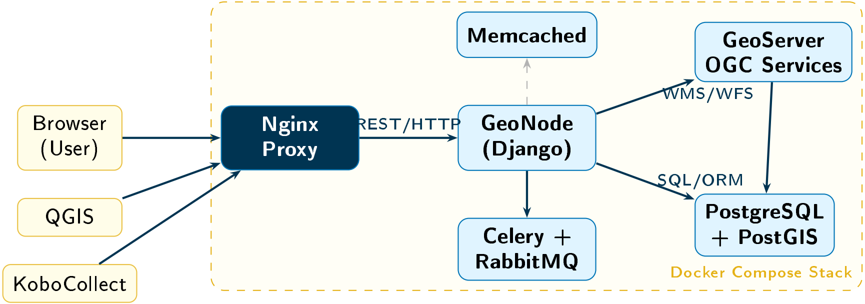
Platform URL

`crd.resilienceacademy.ac.tz`

This Training

All exercises use **real data** from Tanga and Mwanza on the live CRD platform.

CRD Architecture Overview



Component Flow

Browser → Nginx → Django (GeoNode) → GeoServer → PostGIS

Live Demo: Exploring CRD

Exercise: Navigate the Live Platform

Open `crd.resilienceacademy.ac.tz` in your browser and complete these tasks:

1. Browse the **Catalogue** — how many datasets do you see?
2. Search for "**mwanza**" — list the layers you find
3. Search for "**Tanga**" — find `Tanga_imagery_reduced`
4. Open a dataset page — examine the **metadata** (abstract, keywords, bounding box)
5. Click **Map View** — pan and zoom the dataset
6. Find the **Developer** page — note the WMS/WFS endpoint URLs
7. Try the API: visit `/api/v2/datasets/?search=mwanza`

Expected Finds

mwanza_businesses, mwanza_offices, mwanza_education_facilities, Tanga_imagery_reduced

Session 1.1 Summary

Key Takeaways

- ✓ SDI = framework of data, metadata, services, standards, people
- ✓ OGC standards (WMS, WFS, WCS, CSW) enable interoperability
- ✓ CRD is Tanzania's climate risk SDI built on GeoNode
- ✓ 287+ datasets covering Tanga, Mwanza, and more
- ✓ Every layer gets automatic OGC endpoints

Next: Session 1.2

Introduction to Software & Applications

- Docker & Docker Compose
- Git version control
- VS Code, Python, QGIS, Postman, pgAdmin
- Hardware requirements

Questions?

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